Characteristics of Nitric Oxide Formation Rates in Turbulent Nonpremixed Jet Flames

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The formation of oxides of nitrogen, NO_x, in turbulent nonpremixed flames is experimentally analyzed making use of velocity, temperature, and major species concentration measurements in jet propane flames. The analysis considered the contributions of Zeldovich (thermal) and Fenimore (prompt) mechanisms at different locations of attached vertical flames with Reynolds numbers of 20,400 and 37,600. The analysis of NO mass formation rates suggests that prompt mechanism dominates NO formation throughout the flames, with peak formation rates occurring in the rich flame region, at locations r/x approximately equal to 0.08, where r and x are the radial and axial coordinate, respectively. NO concentrations are shown to decrease with Reynolds number, as the gas residence times are reduced, although mass flow rates of NO_x per unit mass flow rate of fuel are independent of Reynolds number. © 1999 by The Combustion Institute

NOMENCLATURE

a	volume coefficient for fuel plus stabilizing hydrogen (nondimensional)
A	area of circular section at station x/D (cm^2)
b	volume coefficient for air (nondimensional)
D	burner internal diameter (cm)
f	mixture fraction (nondimensional); mass fraction of burnt material and total mixture
F	a coefficient defined in Eq. 2c (nondimensional)
G	a coefficient defined in Eq. 2d (nondimensional)
k	a coefficient to account for the number of moles of stabilizing hydrogen (nondimensional)
$\dot{m}$	mass flow rate (g/s)
n	number of moles of gas component per mole of combustion gas mixture (nondimensional)
r	radial coordinate
r/D	axial distance within the flame (nondimensional)
Re	Reynolds number (nondimensional), $Re = \rho_{C_3H_8} D U_0 / \eta_{C_3H_8}$
T	temperature (K)
$\bar{U}$	longitudinal average gas velocity ($cm \cdot s^{-1}$)
V	volume of gas per mole of combustion gas mixture (cm^3)
w	NO production rate ($moles \cdot cm^{-3} \cdot s^{-1}$, per 2 moles of NO)
x	longitudinal coordinate (cm)
x/D	axial distance within the flame (nondimensional)

Greek

ΔA	area element (cm^2)
ρ	density ($g \cdot cm^{-3}$)
η	viscosity ($g \cdot cm^{-1} \cdot s^{-1}$)

Subscripts

0	refers to axial velocity at burner exit
C_3H_8	refers to propane properties
i	denotes each of the points where a set of experimental data was acquired at a particular station x/D
P	prompt
Z	Zeldovich

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INTRODUCTION

It is well known that strategies to reduce NO_x emissions from combustion of fuels containing no bound nitrogen, as is the case of most commercial gaseous fuels, are primarily dependent on thermal (Zeldovich), prompt (Fenimore), and nitrous oxide (N_2O) intermediate mechanisms [1, 2]. It is also known that in nonpremixed combustion of hydrocarbon fuels, the first two mechanisms dominate the process of NO formation, whereas for nonhydrocarbon systems only thermal mechanism is generally active. In general, Zeldovich mechanism is dominant in lean to stoichiometric laminar premixed flames, but Fenimore mechanism can be dominant under conditions where Zeldovich NO is small [3–7].

The complexity of the process and the various questions typically raised in the modification of practical systems, [8], suggest that future emission regulations will probably not be met by existing burners and that costly additional fuel reburning end-of-pipe arrangements may be required, unless improved designs are available. It is, therefore, desirable to understand the relative importance of NO formation mechanisms, including its mean rates of formation in flames, so that the analysis of simple jet flame provides a suitable testing environment.

A typical example of the analysis of NO formation mechanisms has been reported for laminar opposed-flow diffusion flames by Drake and Blint [9]. Based upon experimental and calculated values for CH_4/N_2 flames, the authors concluded that Fenimore mechanism is expected to dominate NO formation in laminar, as in turbulent hydrocarbon diffusion flames. Also, the associated numerical/experimental results of Smooke et al. [5] indicated that prompt production path is the dominant formation route of NO in axisymmetric laminar diffusion flames. Atreya et al. [6], on the analysis of radiating diffusion flames, suggested that Fenimore NO initiation mechanism becomes more important for low-temperature flames. In a recent experimental study, Meunier et al. [7] reported that Fenimore mechanism is the dominant route for NO formation in a turbulent propane diffusion lifted flame, and that residence time is the dominant physical parameter associated with NO_x scaling for the flames analyzed. Moreover, the importance of the interaction between instantaneous fuel trajectories and local flame characteristics on emissions has been demonstrated with a multifuel low-NO_x burner with air and fuel staging [8].

Although comparatively more general information on NO_x production in nonpremixed flames can be found in Turns [10], extended control of this process still requires further analysis, through simple combusting environment as that considered in this work.

This paper discusses results obtained in turbulent nonpremixed attached jet propane flames, with Reynolds numbers in the range of 20,400 to 37,600, evaluating the relative importance of Zeldovich and Fenimore mechanisms on NO formation. The analysis is based on the evaluation of the relative contribution of each of the formation rate mechanisms, utilizing experimental data of flame time-resolved temperature and mean concentration of major chemical species obtained at several radial locations in the flames.

The remainder of the paper includes four sections, as follows. The next section describes flow configuration and experimental methods. Then, the necessary background for analysis of NO_x formation rates is presented and the subsequent section presents and discusses sample results. The last section summarizes the main conclusions of the work.

FLOW CONFIGURATION AND EXPERIMENTAL METHOD

The analysis of NO formation rates in this paper is derived from detailed measurements in attached vertical turbulent jet flames of propane in still air (see [11] and [12] for details). The burner consists of a round nozzle with an exit internal diameter of $D = 5$ mm and boundary conditions typical of a fully developed pipe flow. Commercial grade propane ($C_3 \geq 92\%$) was used as jet fluid in mass flow rates of 0.65 and 1.2 g/s, resulting in jet flames with Reynolds numbers of 20,400 and 37,600, and Richardson numbers of 1.5×10^{-4} and 4.1×10^{-6}. Visible flame length is 1.10 and 1.50 m, respectively $Re = 20{,}400$ and 37,600, and corresponds to

global residence times of 2.16 and 1.25 s, at axial velocity at the burner exit, U_0, of 19 and 33 m/s. The Richardson numbers, Ri, and the gas residence times are similar to those defined by Becker and Liang [13] and Turns and Myhr [14], where $Ri = \pi g \rho_\infty L^3/4U_0\dot{m}_0$ (g = acceleration due to gravity, ρ_∞ = ambient air density, L^3 = visible flame length, $\dot{m}_0$ = mass flux at burner exit). A horizontal large flat plate (600 mm outer diameter) was placed at the tip of the jet during the experiments to assure a pure radial movement of the ambient air entrained at the burner level.

A coaxial flow of H_2 was used to stabilize the flame and to attach it to the burner exit. H_2 flow rates were 0.0053 and 0.0134 g/s, respectively for the two flame conditions considered here, Re = 20,400 and 37,600. These values are compared with those used by [15], and correspond to 0.8 and 1.1% of propane mass flow for Re = 20,400 and 37,600 rate, respectively. These mass flow rates result in velocities 20% of the hydrocarbon boundary condition velocity and have a negligible effect on flow development. Spectrometric measurements at 1870 nm H_2O wavelength show that H_2 combustion does not alter the H_2O signal on the spectrometric signature of propane flame 8 and 12 burner diameters downstream from the burner, respectively for Re = 20,400 and 37,600 [12].

Velocity measurements were obtained with a single component dual-beam laser-Doppler velocimeter based on an argon-ion laser light source, and operating in the backscatter mode. Time-resolved temperature measurements were obtained with bare fine wire thermocouples made of Pt:Pt/13%Rh wires with 0.040 mm diameter. The output of the thermocouple was digitally compensated over two decades and the results showed that the energy-containing frequencies of the flames analyzed were negligible above 750 Hz, which is a value smaller than the frequency response of the system. Results for mean temperature and the variance of temperature fluctuations are accurate within 3% and 20% of the measured values, respectively, as demonstrated by Caldeira-Pires and Heitor [11] and Caldeira-Pires [12], and are not repeated here. These two works present a complete description of this technique, including an assessment of main sources of errors. Namely, Caldeira-Pires and Heitor [11] reported maximum thermocouple radiation heat losses of 6% of the compensated temperature value.

Mean concentrations of CO, CO_2, O_2, NO_x, and unburned hydrocarbon (UHC) were measured by gas sampling and subsequent on-line analysis, using an L-shaped water cooled probe with an orifice 1.5 mm in diameter. The analysis was performed on a wet basis with a flame ionization detector for UHC and a chemiluminescence analyzer for the NO, and on a dry basis with infrared analyzers for O_2, CO_2, and CO, as described by Heitor and Moreira [16]. The results are expected to be close to density-weighted averages, within ±10% of the measured values, and were shown to be independent of the suction velocity, but sampling was not performed for $x/D > 50$, because of probe blockage with particulate soot. The measuring accuracy was assessed through the evaluation of the integral of the mean mixture fraction over the area of each radial profile and from the calculated ratios between the measured and stoichiometric C/H and N/O proportion, which have shown a maximum deviation of 3% from stoichiometry [12].

Figure 1 shows typical results of the jet flame studied herein after. Figures 1a–d present isocontour lines of mean temperature, mean mixture fraction, and mean and fluctuating axial velocity, which depict typical contours of a turbulent nonpremixed attached jet flame. Both flames showed similar values for maximum mean temperature, 1361°C and 1355°C for Re = 20,400 and 37,600. Otherwise, Fig. 2 presents values for instantaneous temperature, ranging up to 1950 K, within 300 K of the stoichiometric adiabatic flame temperature of 2270 K.

Instantaneous temperature was calculated using digitally compensated thermocouples signal. The procedure for the numeric compensation included experimental data on the characteristics of the effect of velocity and temperature on the time constant of the thermocouple and was optimized to guarantee temporal resolution of the algorithm. In this context, Caldeira-Pires and Heitor [11] and Ferrão and Heitor [17] present a detailed description of the numerical compensation procedure and of uncertainty analysis. Moreover, Ferrão and Heitor [17]

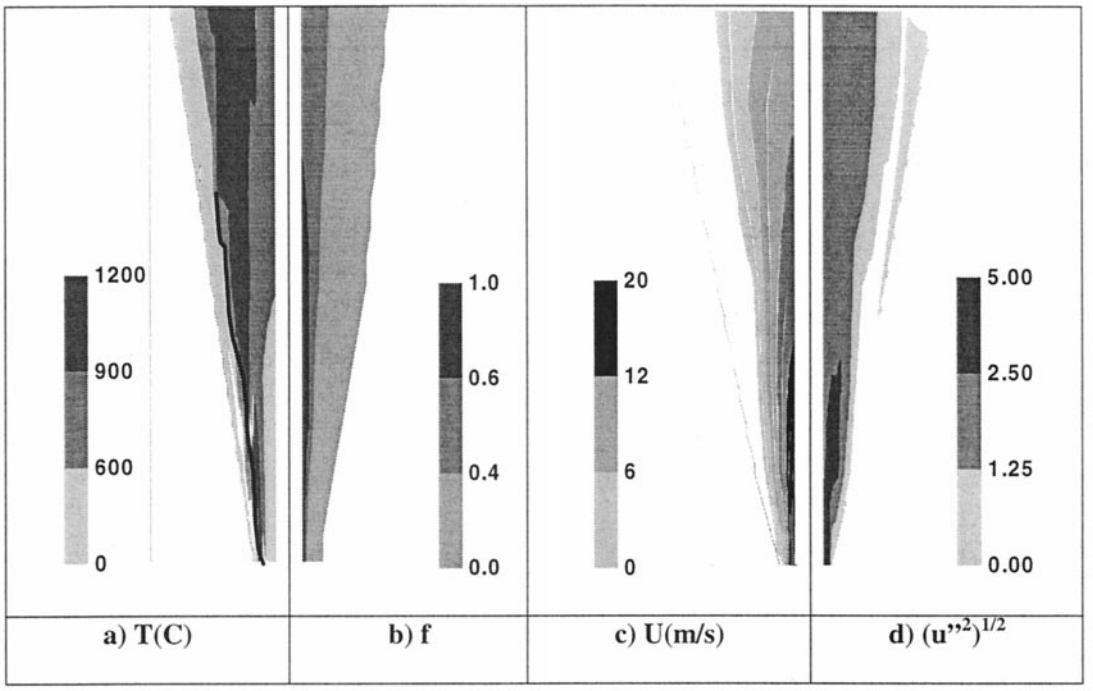

Fig. 1. (a) Isocontours of mean temperature; —— (locus of the stoichiometric mixture fraction, $f = 0.06$). (b) Isocontours of mean mixture fraction. (c) Isocontours of mean axial velocity. (d) Isocontours of rms of the axial velocity.

compared thermocouple-based instantaneous temperature measurements with laser-Rayleigh scattering data. Their analysis of the turbulence nature of strongly recirculating premixed flames showed that both techniques give rise to results in close agreement.

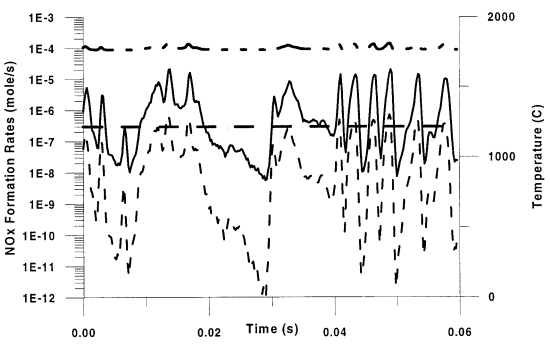

Fig. 2. Time evolution of time-resolved temperature signal and calculated Zeldovich NO_x formation rate; Re = 20,400; $x/D = 20$; $r/D = 2.2$. _____ Temperature; - - - - Wz; – – – $\langle Wz \rangle$; — - - - Wp

BACKGROUND

Assuming that all UHC is effectively C_3H_8 and that air is 21% O_2 and 79% N_2 on a volumetric basis, the combustion reaction can be written as

$$a(C_3H_8 + kH_2) + b(O_2 + 3.76\,N_2) \rightarrow n_{C_3H_8}\,C_3H_8 + n_{O_2}\,O_2 + n_{CO}\,CO + n_{CO_2}\,CO_2 + n_{H_2O}\,H_2O + n_{N_2}\,N_2 + n_{NO_x}\,NO, \quad (1)$$

where n stands for the number of moles of each component per mole of combustion gas mixture, k accounts for the mass flow rate of the stabilizing H_2 ($k = 0.179$ and 0.246, respectively for the flames with $Re = 20{,}400$ and 37,600). If COMP stands for any component of the gas mixture in the flame, then [COMP] = n_{COMP}/V is the concentration of COMP in moles/cm^3, where V corresponds to the volume occupied by 1 mole of mixture at 1 atm ($V = 82.06T$, in cm^3, for temperature T in K, and for approximation of perfect gas).

The integrated mass flow rate of NO over a particular station x/D = constant, one of the parameters discussed in this paper, is estimated by $\dot{m}_{NOx,x/D} = \Sigma_i \rho_{NOx,i}\bar{U}_i\Delta A_i$. In this equation, the subscript i represents the points where temperature, concentration, and velocity measurements were taken at the particular axial station x/D, $\bar{U}_i$ is the longitudinal average gas velocity, and ΔA_i is an area element. This element was either a circle, if $r_i/D = 0$, or a ring, if $r_i/D > 0$, the limits of which were given by $(r_{i+1} + r_i)/2$ and $(r_i + r_{i-1})/2$. The density of NO is provided, in g/cm^3 units, with $\rho_{NOx,i} = 0.366 n_{NOx,i}/T$.

The analysis of NO formation rates is achieved making use of the equations derived by Røkke et al. [18, 19] for methane, propane, and natural gas flames. Their theory employs reduced chemical-kinetic mechanisms for the reaction-sheet regime along with simplified mass, momentum, and mixture-fraction balances for the turbulent flame height. Their explicit expression for local NO production rates, as derived from instantaneous Zeldovich, w_Z, and prompt NO, w_P, formation processes, is given as follows, in terms of molar rates of 2 NO per unit volume:

$$w_Z = 9 \times 10^{12} T^{0.3} \exp\left(\frac{-38{,}440}{T}\right)[N_2][O_2], \tag{2a}$$

$$w_P = \left\{\frac{3 \times 10^{10} \exp\left(\frac{-2900}{T}\right) F^{3/2} G [N_2][CH_4][O_2]^{3/2}[H_2O]^{1/2}}{([M] - [N_2])([M] - [N_2] - [\bar{C}])} + \frac{4 \times 10^{9} \exp\left(-\frac{2450}{T}\right) F [N_2][C_3H_8][O_2][H_2O]}{([M] - [N_2])([M] - [N_2] - [\bar{C}])}\right\}\left[1 + 3000 \exp\left(-\frac{15{,}185}{T}\right)\right], \tag{2b}$$

where T is temperature (K), M stands for any molecule, $\bar{C}$ stands for any combustion product component containing carbon atoms (approximated by CO_2 [18]), [ ] denotes molar concentration (mole/cm^{3}s), and

$$F = 1 - 1.1 \times 10^{-10} T^3 \exp\left(-\frac{4051}{T}\right)\frac{[CH_4]}{[O_2]} - 0.9 \exp\left(-\frac{4243}{T}\right)\frac{[C_3H_8]}{[H_2O]}, \tag{2c}$$

$$G = 1 + 40 \exp\left(\frac{-5900}{T}\right) + 2.6 \times 10^{13} T^{-4} \exp\left(-\frac{7600}{T}\right)\frac{[CH_4]}{[H_2O]}. \tag{2d}$$

In the present case, we assume the gas species include H_2O, O_2, CO_2, CO, UHC = C_3H_8, NO, and N_2. Therefore, $[CH_4] = 0$, $F = 1 - 0.9 \exp(-4243/T)[C_3H_8]/[H_2O]$, and $G = 1 + 40 \exp(-5900/T)$. Because in the present case $[CH_4] = 0$, G is not necessary in equation (2b).

Figure 2 depicts the higher dependence on temperature of Zeldovich mechanism, w_Z, using constant mean values of the concentration of the chemical species. An overall increase of 200 degrees in instantaneous temperature would augment mean w_Z less than 10 times. The same figure shows that w_P results are more than two orders of magnitude higher than mean w_Z value. Mean values of NO_x formation rates were averaged using a population of 5000 calculated points, based on temporal series of compensated temperature signals.

As stated by Røkke et al. [18], uncertainties are about a factor of 2 in the Zeldovich formation rate and a factor of 10 in the prompt rate. Nevertheless, the formulas above can provide reasonable estimates and yield useful conclusions regarding the relative contribution of each type of NO formation process in the flame. In addition, due to the larger activation energy, NO production due to the Zeldovich mechanism is the slower process [20].

RESULTS AND DISCUSSION

Figure 3 presents the variation of NO concentrations and Zeldovich and prompt NO forma-

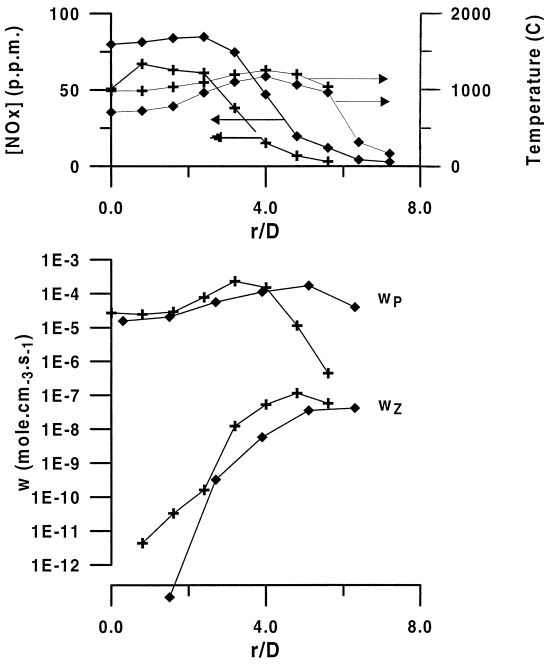

Fig. 3. Radial profiles of NO concentration, mean temperature, and rates of formation of Zeldovich and prompt NO; $x/D = 50$; ◆ $Re = 20{,}400$; ✚ $Re = 37{,}600$.

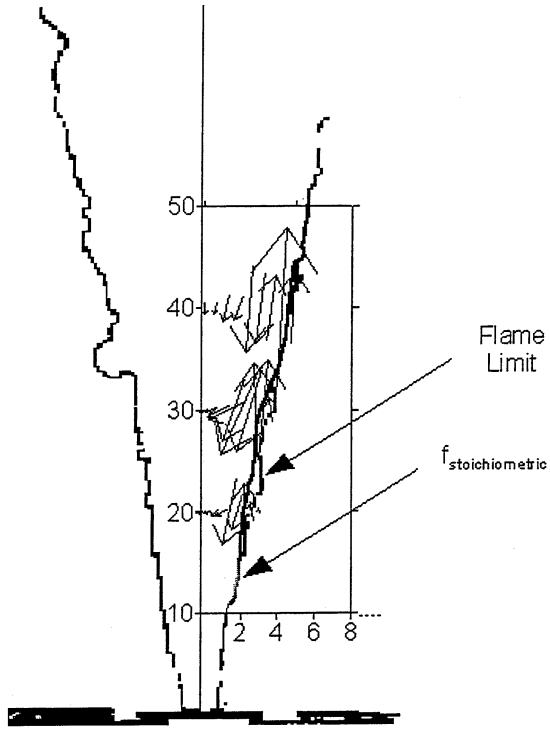

Fig. 4. Turbulent heat flux vectors distribution (superposed to the digitized flame image contours); $Re = 20{,}400$; $x/D = 20$, 30, and 40.

tion rates as functions of the normalized radial direction r/D, for the station located at 50 diameters and both flame Reynolds numbers of 20,400 and 37,600. Whereas the NO concentrations come from direct measurements, the formation rates were computed using Eq. 2 taking into account experimental values of time-resolved temperature and mean concentration of chemical species. Besides, as was also highlighted by one of the referees of this work, the turbulent diffusion flames here studied typically exhibit high variance of the probability density functions (pdf) of most of their quantities, and therefore feature strong turbulence–chemistry interactions.

The pdf distribution of the temperature (not shown here) shows a non-Gaussian behavior, typical of nonpremixed jet flames. Temperature skewness, $S(t)$ (which indicates the probabilistic asymmetry of the measured variable), changes signal at shear layer position characterized by positive values throughout the radial profiles, except at a narrow zone outside maximum temperature zone due to the entrainment of external cold air into hot combustion gases. Measurements of $S(u)$ and $S(t)$ indicate that u'' and t'' are skewed in the opposite directions.

The measurements show that NO formation rates peak near the maximum temperature, due to the dependence of NO formation rates on temperature, whereas the NO concentrations reach maximum values in the inner region of the flame, due to the gases' diffusion process as combustion flow proceeds downstream.

In a former paper [12], the characteristics of turbulent heat transport in nonpremixed jet flames are analyzed, addressing the effects on mixing patterns due to flame chemistry–turbulence transport interactions. Figure 4 presented some of the main results achieved, and depicted that the correlation between radial velocity and temperature is always smaller than those between axial velocity and temperature, suggesting that turbulent motions are more efficient in transporting heat along the axial direction, toward the center axis of the flame. As suggested by Hardalupas et al. [21], the behavior of turbulent heat transport can be extended to the

conserved scalar. Basically, this assumption depicted preferential paths whereas chemical species at high temperatures are moving together, allowing the continuation of the high activation energy reactions characteristics of NO_x formation. This explains why the highest NO_x concentration is measured in a position inner than where the peak of NO_x formation is calculated.

At fuel-lean conditions, where temperature drops below the levels of significant NO production, the correlation of NO and temperature also becomes weaker due to the moderating effect of the mixing process as already mentioned [22]. It is also observed that, at the same axial station, the NO concentration curve for $Re = 20{,}400$ is higher compared to the NO curve corresponding to $Re = 37{,}600$, as a consequence of lower gas velocities and higher gas residence times for the first.

Figure 3 also shows that maximum values of Zeldovich NO_x formation rate are up to three orders of magnitude lower than those calculated to prompt NO_x, even including uncertainties stated by Røkke et al. [18]. These results clearly indicate that the prompt production path is the dominant formation route of NO_x in the jet flames studied herein, whereas the Zeldovich NO_x mechanism plays a minor role in total NO production. This fact was already mentioned in the literature for laminar diffusion flames [5, 6] and turbulent lifted jet flames [7].

The evolution of the temperature and of the molar concentrations of C_3H_8, O_2, N_2, and H_2O, for $x/D = 50$ and $Re = 20{,}400$ and 37,600 was taken from values reported by [12]. These chemical species are responsible for prompt NO production. The measurements depict that the behavior of these parameters, at the same axial station, is similar for both Reynolds numbers, as previously reported for turbulent jets by, e.g. [23, 24]. Near the center line ($r/D = 0$), temperature is still relatively high (50% of the maximum value measured) and the gas contains fuel (40%), entrained N_2 (40%), and some entrained unreacted O_2 (<1%), such that prompt NO can be still formed there. As pointed out by [22], for each 100 K of variation in the usual temperature range the Zeldovich formation rate varies by a factor ranging from 5 to 2, and the Fenimore by a factor from 0.2 to 0.4.

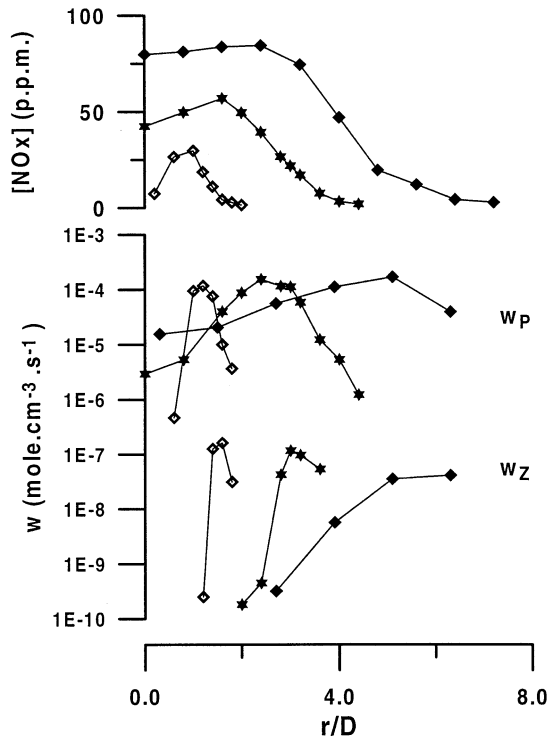

Fig. 5. Radial profiles of NO concentration and rates of formation of Zeldovich and prompt NO; $Re = 20{,}400$; ◇ $x/D = 10$; ★ $x/D = 30$; ◆ $x/D = 50$.

Figure 5 shows that maximum values of NO_x prompt formation rate, w_P, are reached at $r/D = 1.2$, 2.4, and 4, respectively for $x/D = 10$, 20, and 30. These values are smaller than those for the maximum temperature position at each axial station, and reflect the relative influence of the chemical species concentration on the formation rate. Maximum w_P calculated presents similar values, namely 10^{-4} mole/($cm^3 \cdot s$), for all the axial stations, and for both Reynolds conditions investigated in this work.

Moreover, the calculated maximum value of w_P is located at $r/x = 0.08$ for both Reynolds numbers and all axial stations investigated, which is a characteristic value to represent 50% maximum velocity radial position/axial position ratio of nonreacting turbulent jets. This position can be represented by a cone centered at the burner tip, with a vertices semi-angle of 4.6°. On the other hand, stoichiometric mixture ratio ($f_{stoich} = 0.063$) is calculated to occur at $r/x = 0.11$, which can also characterize a cone with vertices semi-angle of 6.3°. Therefore, peak w_P

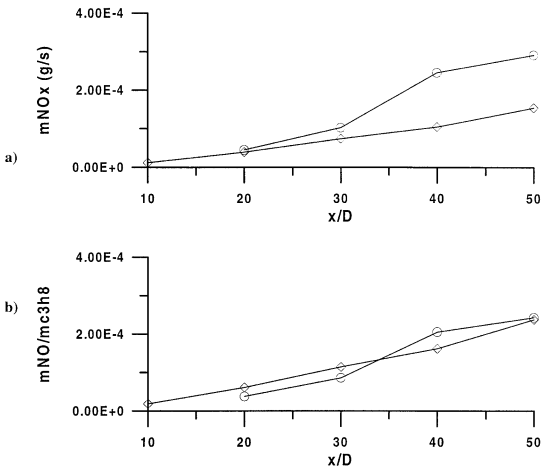

Fig. 6. (a) Axial profile of the calculated mass flow rate of NO; (b) axial profile of the ratio between the calculated volume flow rates of NO and C_3H_8; ◇ Re = 20,400 and ○ 37,600.

reflects the dependence of prompt NO_x formation rate with the rich region of the flame.

The analysis in the present work shows that prompt formation rates, w_P, as given by the formulation of Røkke et al. [18], were in general three orders of magnitude higher than Zeldovich's, w_Z, even considering the mentioned maximum uncertainty level of 20 times in Eq. 2. The maximum temperature measured throughout all points was 1635 K (at x/D = 20 and r/D = 2.2, for Re = 20,400), which is still low to produce other than prompt NO. The minimum value for the w_P/w_N ratio is 1.8, for Re = 20,400, x/D = 20, r/D = 2.8. Besides, at this particular point w_P was two orders of magnitude lower than the maximum value attained at the same station in the flame.

Figure 6 shows the evolution of the calculated mass flow rates of NO, and of the ratio between the calculated mass flow rates of NO and fuel, as functions of the axial coordinate x/D and for both Reynolds conditions. It is observed (Fig. 6a) that the mass flow rate of NO increases as the flow progresses upwards due to pollutant formation, being larger for the larger Reynolds number, simply because in this case there are larger quantities of fuel to react. However, if the mass flow rate of NO is normalized by the mass flow rate of fuel (i.e., by the flame power output), the results are similar for the two flow conditions, as shown in Fig. 6b. This means that turbulence, at least up to x/D = 50, does not work towards producing a decrease on NO_x generation. Above x/D = 50, temperatures were observed to decrease significantly, being the same trend expected for fuel concentration and, therefore, for the NO generation rates.

CONCLUSIONS

The characteristics of NO_x formation are analyzed in propane nonpremixed turbulent jet flames, for Reynolds numbers of 20,400 and 37,600. The main conclusions may be summarized as follows.

Based on the formulae presented by Røkke et al. [18, 19], there is no relevant source of NO other than that based on Fenimore's mechanism in the flame regions investigated here.

The absolute temperature levels, together with appropriate combination of UHC, N_2, O_2, and H_2O concentrations, are responsible for prompt NO formation. The evidence is derived from the fact that near the longitudinal axis of the flame and at $x/D \geq 20$, temperature is still relatively high (>700°C) and there exist entrained N_2 and some entrained unreacted O_2, such that prompt NO_x is formed.

For each longitudinal station, NO formation rates are maxima near the local temperature peaks, but NO_x concentrations are maxima for smaller radii, due to turbulent transport toward the center of the flame as the flow proceeds downstream.

At similar axial stations in the flame, NO concentrations increase by decreasing the Reynolds number, due to increased gas residence times. Also, NO mass flow rates increase with the Reynolds number, although for a given axial station in the flame, mass flow rates of NO_x per unit mass flow rate of fuel are independent of the Reynolds number.

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